$2 \qquad 2r$

4 (a) The internal energy of an ideal gas is the total kinetic energy of all the particles (atoms and/or molecules) in the gas due to their random motion. There is no microscopic potential energy since the particles experience no inter-molecular forces of attraction.

(b) (i) Using $pV = nRT$

$$n = \frac{pV}{RT} = \frac{\left(1.0 \times 10^5\right)(0.075)}{(8.31)(273.15 + 25)} = 3.027 \text{ mol}$$

$\therefore$ mass of air $= 3.027 \times 0.030 = 0.0908$ kg

(ii) Since mass of air $\propto$ number of mole of air, i.e. $m \propto n$
Using $pV = nRT$

$$\frac{n}{V} = \frac{p}{RT}$$

$$\because \frac{m}{V} \propto \frac{n}{V}$$

$$\therefore \frac{m}{V} \propto \frac{p}{RT} \Rightarrow \rho \propto \frac{p}{RT}$$

$$\therefore \quad \frac{\rho_{25^\circ\text{C}}}{\rho_{200^\circ\text{C}}} = \frac{T_{200^\circ\text{C}}}{T_{25^\circ\text{C}}} = \frac{200 + 273.15}{25 + 273.5} = 1.59 \quad (p \text{ is constant})$$